

Engineering Mechanics: ME101

Lecture 1: Statics



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Pre Mid-Semester Portion (Statics)

S.No	Topics	Total Lectures
1	Basic principles: Equivalent force system, Equations of equilibrium, Free body diagram, Reaction, Static indeterminacy;	4
2	Structures: Difference between trusses, frames and beams, Assumptions followed in the analysis of structures; 2D truss; Method of joints; Method of section;	3
3	Frame; Simple beam; types of loading and supports; Shear Force and bending Moment diagram in beams; Relation among load, shear force and bending moment	4
4	Friction: Dry friction; Description and applications of friction in wedges, thrust bearing (disk friction), belt, screw, journal bearing (Axle friction); Rolling resistance	3
5	Virtual work and Energy method: Virtual Displacement; Principle of virtual work; Applications of virtual work principle to machines; Mechanical efficiency; Work of a force/couple (springs etc.); Potential energy and equilibrium	2
6	Radius of gyration; Parallel axis theorem; Product of inertia, Rotation of axes and principal moment of inertia; Moment of inertia of simple and composite bodies. Mass moment of inertia	3

Post Mid-Semester Portion (Dynamics)

S.No	Topics	Total Lectures
8	Kinematics of Particles: Rectilinear motion; Curvilinear motion; Use of Cartesian, polar and spherical coordinate system; Relative and constrained motion; Space curvilinear motion	5
9	Kinetics of Particles: Force, mass and acceleration; Work and energy; Impulse and Momentum; Kinematics of System of Particles	5
10	Kinematics and Kinetics of Rigid Bodies: Translation; Fixed axis rotational;	2
11	General plane motion; Coriolis acceleration; Work energy; Power; Potential energy	3
12	Impulse-momentum and associated conservation principles;	2
14	Euler equations of motion and its application.	3

Text and references

□ Textbooks:

- 1) I. H. Shames, Engineering Mechanics: Statics and Dynamics, 4th Ed., PHI, 2002.
- 2) F. P. Beer and E. R. Johnston, Vector Mechanics for Engineers, Vol I - Statics, Vol II – Dynamics, 3rd Ed., Tata McGraw Hill, 2000.

□ References:

- 1) J. L. Meriam and L. G. Kraige, Engineering Mechanics, Vol I – Statics, Vol II – Dynamics, 5th Ed., John Wiley, 2002.
- 2) R. C. Hibbler, Engineering Mechanics, Vols. I and II, Pearson Press, 2002.
- 3) D Gross, W Hauger et al., Engineering Mechanics 1,11, Springer 2009

Grading Policy

S.No	Type	Weightage
1	Tutorial & Home Assignments	10%
2	2 Quizzes	20%
3	Mid-Semester Exam	30%
4	End-Semester Exam	40%

❑ Note on Tutorial and Assignments weightage

- ✓ There will be 11 tutorials, each of 1% weightage. Best 10 out of 11 will be considered.

Lecture 1: Statics

INTRODUCTION

❑ The Engineering science is usually divided into Four main of number topics such as:

1. Solid Mechanics

2. Fluid Mechanics

3. Heat Transfer

4. Properties of materials

✓ There are close links between them in terms of the physical principles involved and methods of analysis employed.

❑ **Definition of Solid Mechanics:**

✓ The solid mechanics is a subject may be defined as a branch of applied mechanics that deals with behaviors of **solid bodies** subjected to **various types of loadings**.

❑ Solid Mechanics is a collection of physical laws, mathematical techniques and computer algorithms that can be used to predict the behavior of a solid material that is subjected to mechanical or thermal loading.

STREAMS OF SOLID MECHANICS

❑ **Solid Mechanics usually subdivided into two streams**

1. Mechanics of rigid bodies or simply Mechanics and

2. Mechanics of deformable solids.

✓ Mechanics of deformable solids which is branch of applied mechanics that is known by several names i.e. **Strength of materials, Mechanics of materials** etc.

❑ **Mechanics of rigid bodies:**

✓ The mechanics of rigid bodies is primarily concerned with the **static and dynamic behavior under external forces** of engineering components and systems which are treated as substantially **very strong and undeformable**.

✓ Primarily here we deal with the different **forces and motions** associated with particles and rigid bodies.

Mechanics

- ❑ Oldest and the most highly developed branch of physics.
- ❑ The task of mechanics include the description and determination of the motion of bodies as well as the investigation of the forces associated with the motion.
- ✓ **Analytical Mechanics:** Analytical methods of mathematics are applied with the aim of gaining principle insight into the laws of mechanics.
- ✓ **Engineering Mechanics:** Concentrate on the needs of the practicing engineer, engineer has to analyze bridges, building, machines, vehicles or components of microsystem to determine whether they are able to sustain certain loads or perform certain movements.

Classification of Mechanics

❑ Based on the state of material (2 categories):

- ✓ **Mechanics of Solids**
- ✓ **Hydrodynamic or gas dynamics**

❑ Based on state of rest or motion:

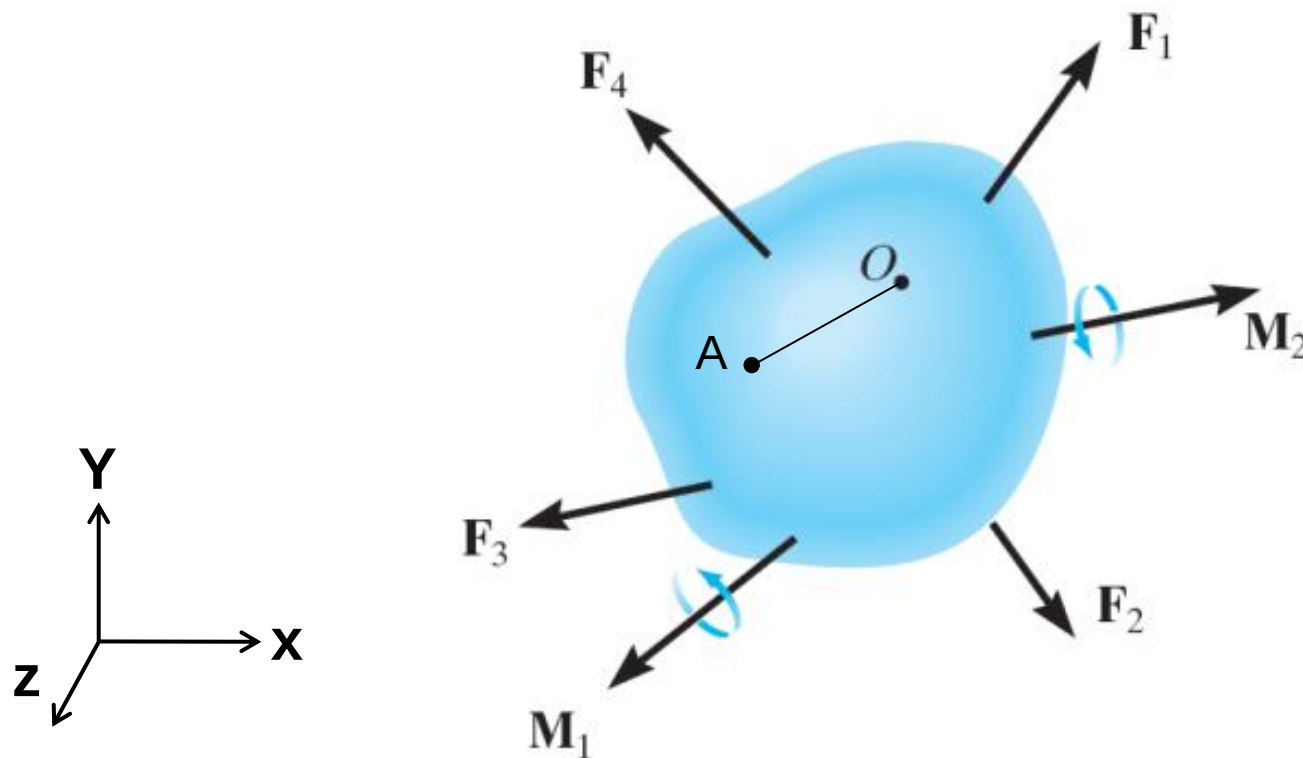
- ✓ **Statics** (Latin: status=standing) deals with the equilibrium of bodies subjected to forces.
- ✓ **Dynamics** (Greek: dynamic =force) is subdivided into **kinematics** and **kinetics**.
 - **Kinematics** (Greek: kiness = movement) **investigates the motion of bodies** without referring to forces as a cause or result of the motion. (ex. displacement, time, velocity, etc.)
 - **Kinetics**: The **study of forces that cause motion** (ex. torque, gravity, friction, etc.).

Applications of Solid Mechanics

1. **Civil engineering** : designing foundations or structures;
2. **Mechanical engineering** : designing load bearing components for vehicles; power generation and transmission;
3. **Manufacturing engineering**: designing metal and polymer forming processes; machining etc;
4. **Geomechanics**: modeling the shape of planets; tectonics; and earthquake prediction;
5. **Biomechanics** : designing implants; bone mechanics; as well as modeling stress driven phenomena controlling celular and molecular processes;
6. **Materials Science** : designing composites; alloy microstructures, thin films, and developing materials processing;
7. **Microelectronics** : designing failure resistant packaging and interconnects;
8. **Nanotechnology**: stress driven self-assembly on surfaces; manufacturing processes such as nano-imprinting; modeling atomic-force microscope/sample interactions.

Rigid body mechanics :: Basics

- Rigid Body: **No deformation** under any load
 - *Change in distance* between any two points *negligible* as compared to body dimensions



What is Rigid body?

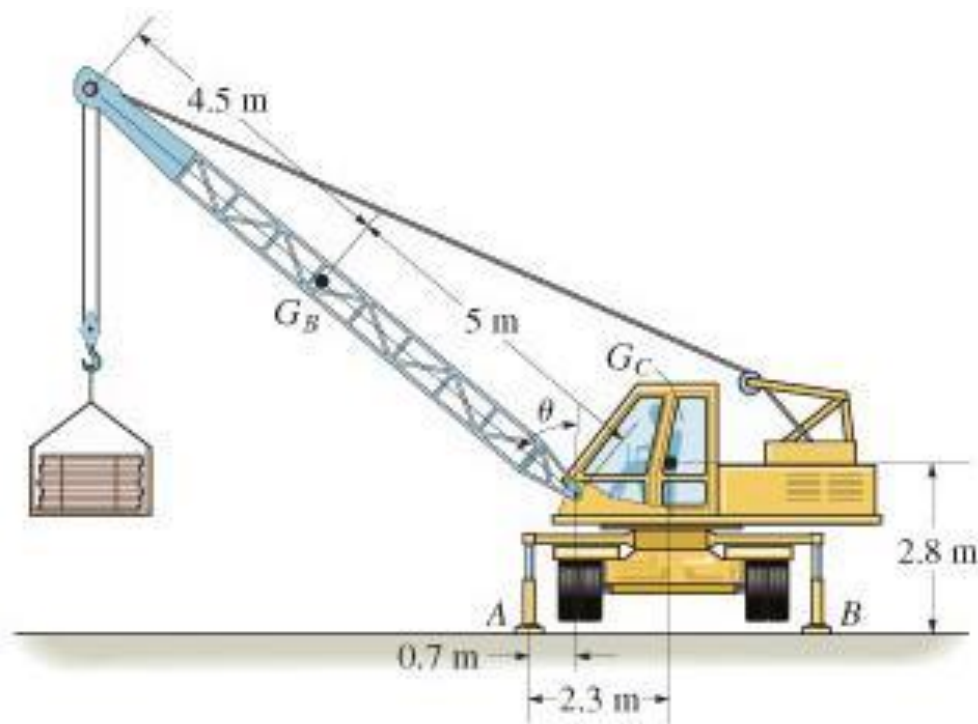
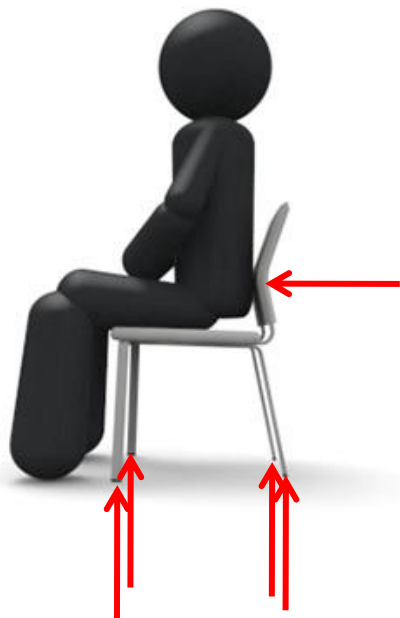
- **Rigid Body**

- A **combination** of large number of **particles** in which all particles remain at a **fixed distance** (*practically*) from one another before and **after** applying a load
- **Material properties** not required when analyzing the forces acting on the body
- Design and analysis of many types of structural members, mechanical components, electrical devices, etc., encountered in engineering.

Rigid body mechanics :: Statics

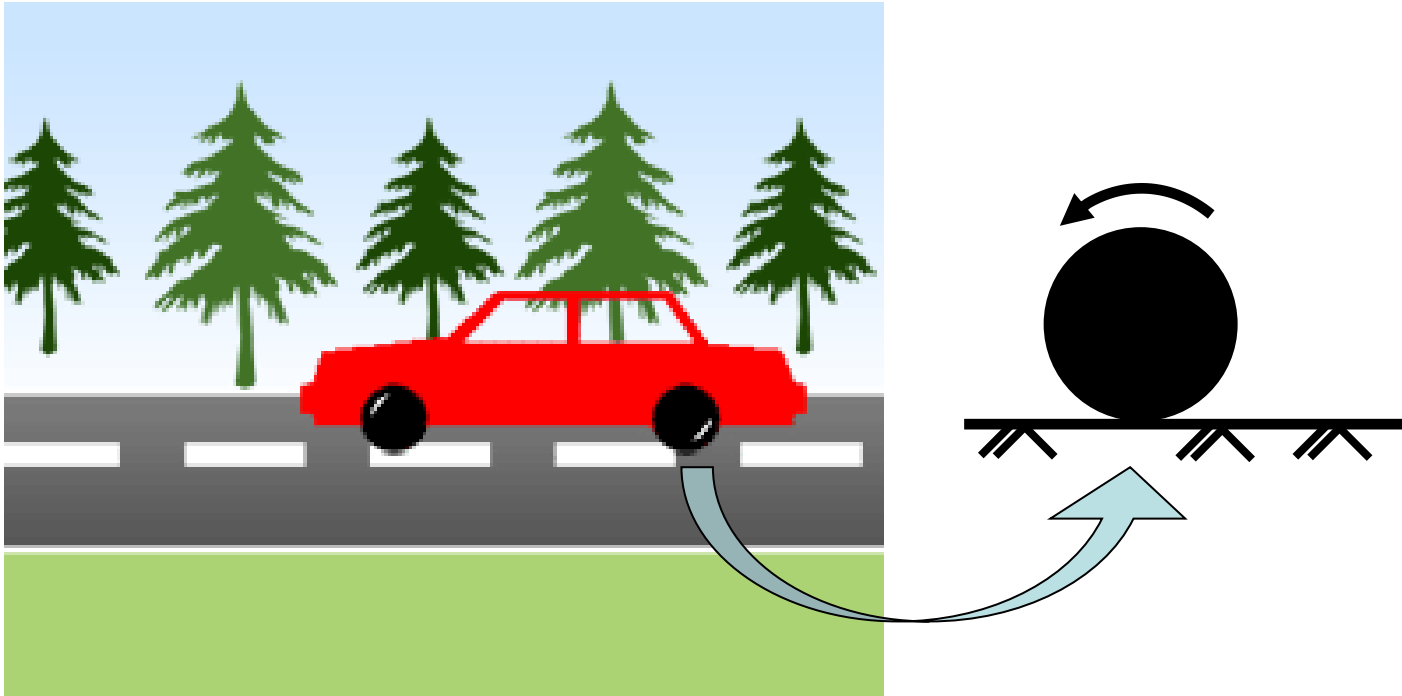
- **Statics**

- **equilibrium** of rigid body under **action of forces**



Rigid body mechanics :: Dynamics

- **Dynamics**
 - **motion** of bodies (*acceleration/deceleration*)



Mechanics: Fundamental Concepts

- **Length (Space)**
 - **Position** of a point in space
 - Coordinate system
 - **Cartesian** (x, y, z)
 - **Spherical** (r, θ, ϕ)
 - **Cylindrical** (ρ, ϕ, z)
 - Describe **size** of the **physical system**
 - Dimensions
 - Distance, geometric properties
 - **Basic quantity/dimension**

Mechanics: Fundamental Concepts

❑ The basic concepts used in mechanics are as follows:

- **Length (Space):** It is needed to locate position of a point in space, & describe size of the physical system: Distances, Geometric Properties.
- **Time:** It is the measure of succession of events and it is basic quantity in Dynamics. To define an event, it is not sufficient to indicate the its position in space. The time of the event should also be given.
- **Mass:** It is the quantity of matter in a body and it is the measure of inertia of a body (its resistance to change in velocity).
- **Force:** It represents the action of one body on another. It is characterized by its magnitude, direction of its action, and its point of application. It can be exerted by actual contact or at a distance (e.g. gravitational, electrical and magnetic forces). **Force is a Vector quantity.**

Mechanics: Fundamental Concepts

- **Force**

- Tends to move a body along its direction
 - **Change in velocity**
- Characterization
 - **Magnitude**
 - **Direction**
 - **Point of application**
- **Derived quantity (MLT^{-2})**

} **Vector**

Mechanics: Fundamental Concepts

- More about **mass** and **weight**
 - **No change in mass with change in location of body**
 - **Weight** refers to the gravitational attraction of the earth on a body or quantity of mass. Its magnitude depends upon the elevation at which the mass is located.
- ❖ **Weight of a body is the gravitational force acting on it**

Concepts of Units

- ❑ **Fundamental units:** The measurements of physical quantities is one of the most important operation in engineering. Every quantity is measured in terms of some arbitrary, but internationally accepted units, called **fundamental units**. There are 3 fundamental quantities:
 1. **Length**
 2. **Mass and**
 3. **Time**
- ❑ **Derived Units:** Sometimes, physical quantities are expressed in other units, which are derived from fundamental units, known as derived units, e.g., units of area, pressure, velocity etc.
- ❑ **System of Units:** There are only **4 system of units**, which are commonly used i.e.:
 1. C.G.S (i.e. cm, gm and second)
 2. F.P.S (i.e. ft, pound and second)
 3. M.K.S (i.e. meter, kg and second) and
 4. **S.I. units (i.e. meter, kg and second) we used**

Relationship and Difference between M.K.S and S.I. units

- ❑ Both M.K.S and S.I. units have same **fundamental units**.
- ❑ But there is a slight variation in their **derived units**.
- ❑ The SI system standardized on the MKS system and added Kelvin or Celsius, ampere, mole and luminous intensity i.e. candela to the three basic units.

❖ **The following derived units generally we will used in this course work:**

Density (mass density)	kg/m^3
Force	N
Pressure	N/m^2
Work done	$\text{J}=\text{N}\cdot\text{m}$
Power	$\text{W}=\text{J/s}$
Kelvin	K

Some standard abbreviations of units

Units	Wrong	Correct
Metre or Metres	M	m
Kilometer or Kilometers	KM/Km	km
Kilogram or Kilograms	KG/Kg	kg
Second or Seconds	S	s
Newton	n	N
Work done or Newton x metres	nM/NM	N-m
Kilonewton x metres	KNm/	kN-m
Centigrade	C	oC
Kelvin	k	K

Mechanics: Units

Four Fundamental Quantities

Quantity	Dimensional Symbol	SI UNIT	
		Unit	Symbol
Mass	M	Kilogram	kg
Length	L	Meter	m
Time	T	Second	s
Force	F	Newton	N

Basic Unit

$$F = ma$$

$$\rightarrow N = \text{kg.m/s}^2$$

1 Newton is the force required to give a mass of 1 kg an acceleration of 1 m/s²

$$W = mg$$

$$\rightarrow N = \text{kg.m/s}^2$$

Mechanics: Units Prefixes

	Exponential Form	Prefix	SI Symbol
<i>Multiple</i>			
1 000 000 000	10^9	giga	G
1 000 000	10^6	mega	M
1 000	10^3	kilo	k
<i>Submultiple</i>			
0.001	10^{-3}	milli	m
0.000 001	10^{-6}	micro	μ
0.000 000 001	10^{-9}	nano	n

Mechanics: Fundamental Concepts

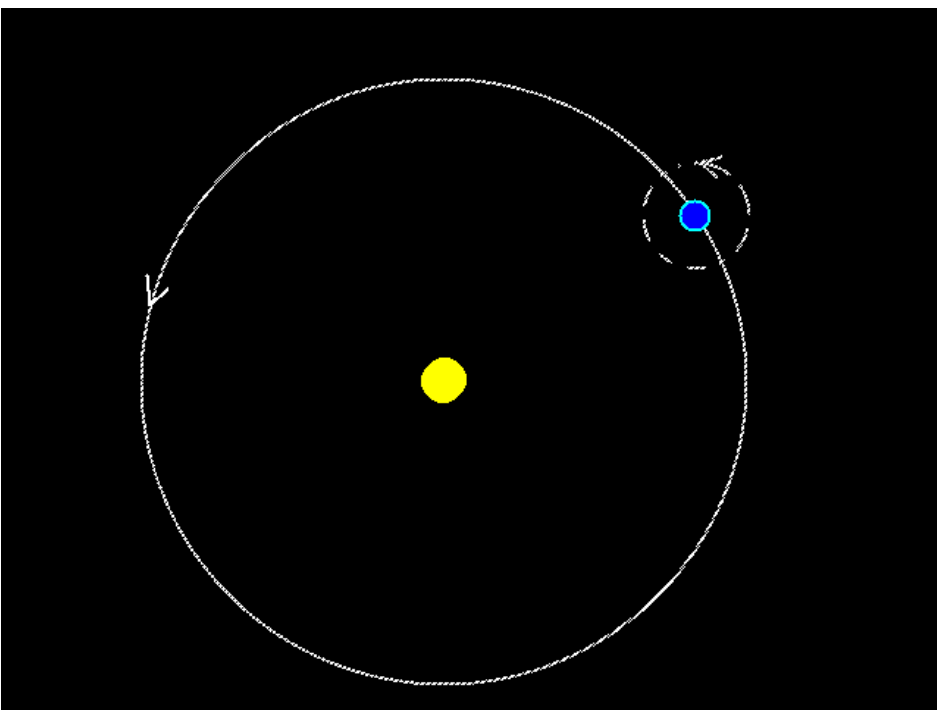
□ Newtonian Mechanics

- Length, Time, and Mass are absolute concepts independent of each other. (this is not true in relativistic mechanics, where the time of an event depends upon its positions and the mass of the body varies with its velocity.)
- Force is a derived concept is not independent of the other three fundamental concepts.
- ✓ Force acting on a body is related to the mass of the body and the variation of its velocity with time.

Mechanics: Idealization as particle

- **Particle**

- A body with **mass** but with **negligible dimensions**. It is a very small amount of matter which may be assumed to occupy a single point in space



: Size of earth insignificant compared to the size of its orbit

: Earth can be modeled as a particle when studying its orbital motion

: **Simplified analysis - geometry of the body is not involved in the analysis.**

Mechanics: Idealizations

□ Force:

- The concept of force can be taken from our daily experience. Although forces cannot be seen or directly observed, we are familiar with their effect.
- The direction of the force can be described by its line of action and its sense of direction (orientation).
- ❖ A quantity determined by magnitude and direction is called a vector. In contrast to a free vector, which can be moved arbitrarily in space provided it maintains its direction, a force is tied to its line of action and has a point of application. Therefore **the 'Force' is a bound vector.**

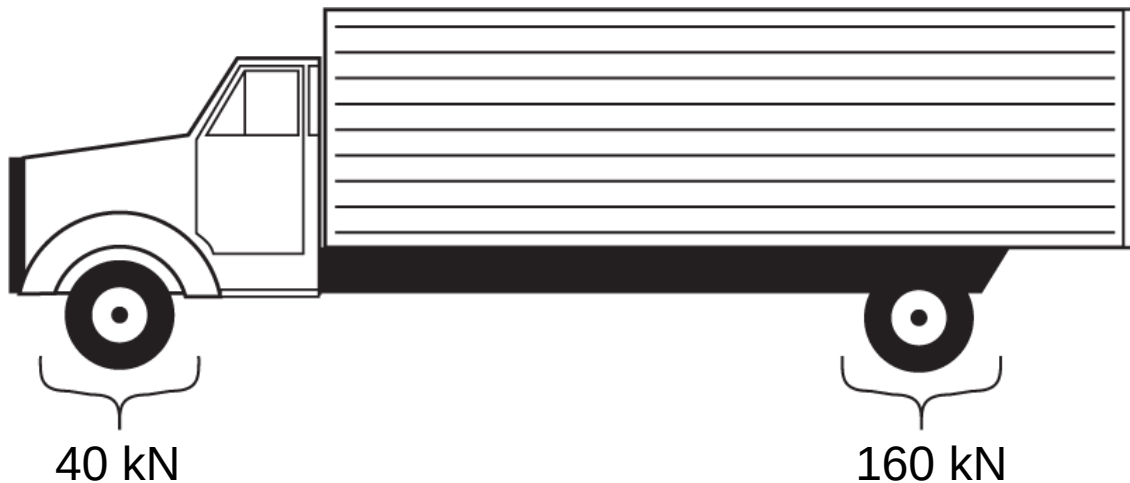
Classification of forces

- **Concentrated force:** A single force with a line of action and a point of application.
- **Volume force:** It is a force that is distributed over volume of the a body or portion.
- **Area force:** It occurs in the region where two bodies are in contact.
- **Active forces:** Physically prescribed forces in a mechanical system
- **Reactive force:** Generated if the freedom of movement of a body is constraint.

Mechanics: Force idealization

- **Concentrated Force**

- **Line of action of weight** through the **centre of gravity** of the body
- **Area over which the load is applied is very small** compared to the **overall size** of the body



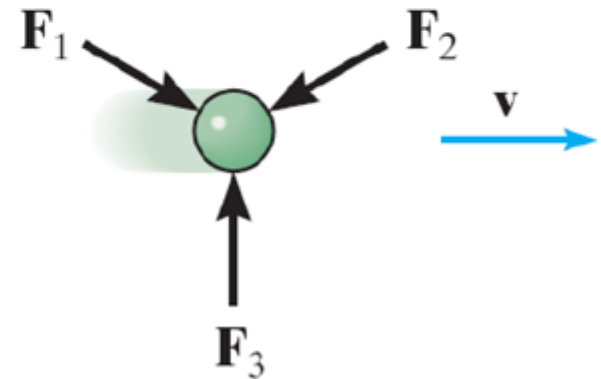
Ex: Contact Force
between a wheel
and ground.

Mechanics: Newton's Three Laws of Motion

Basis of formulation of rigid body mechanics.

First Law: A particle originally at rest, or moving in a straight line with constant velocity, tends to remain in this state provided the particle is not subjected to an unbalanced force.

First law contains the principle of the equilibrium of forces → main topic of concern in Statics



□ It provides the Principle of force equilibrium (statics)

Mechanics: Newton's Three Laws of Motion

- **Basis of rigid body dynamics**

- **Second Law:** A particle of mass “ m ” acted upon by an unbalanced force “ F ” experiences an acceleration “ a ” that has the same direction as the force and a magnitude that is directly proportional to the force

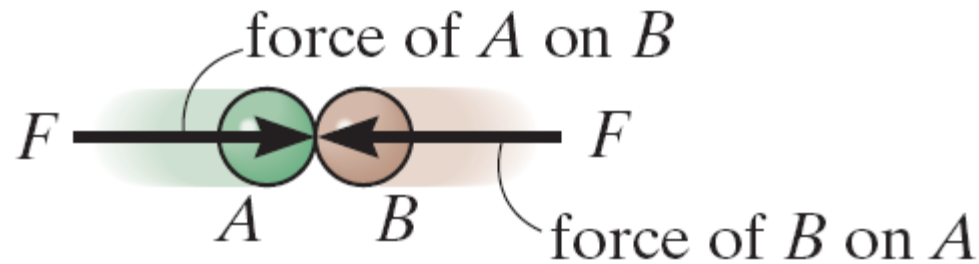


Accelerated motion

- ❖ It forms the basis of most of the analysis of dynamics

Mechanics: Newton's Three Laws of Motion

- Application in both statics and dynamics
 - **Third Law**: The mutual forces of action and reaction between two particles are equal, opposite and collinear.



Action – reaction

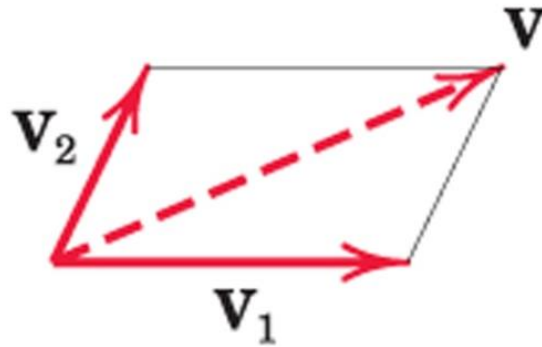
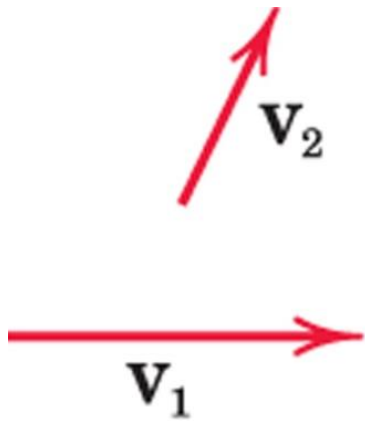
- Third law is basic to our understanding of Force → Forces always occur in pairs of equal and opposite forces.

Mechanics: Scalars and Vectors

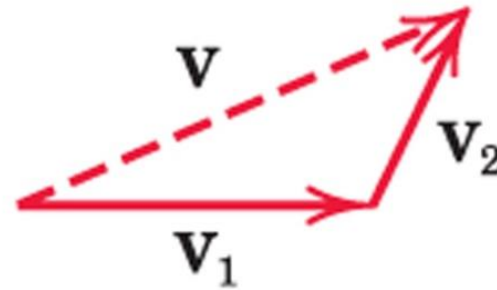
- Scalar
 - Only **magnitude** is associated with it
 - *e.g.*, time, volume, density, speed, energy, mass etc.
- Vector
 - Possess **direction** as well as **magnitude**
 - Parallelogram law of addition (and the triangle law)
 - *e.g.*, displacement, velocity, acceleration etc.
- Tensor (it depends on plane of orientation also)
 - *e.g.*, stress (3×3 components)

Mechanics: Scalars and Vectors

- Laws of vector addition
 - Equivalent vector $\mathbf{v} = \mathbf{v}_1 + \mathbf{v}_2$ (Vector Sum)



✓ Parallelogram law
of vector addition



✓ Triangle law of vector
addition

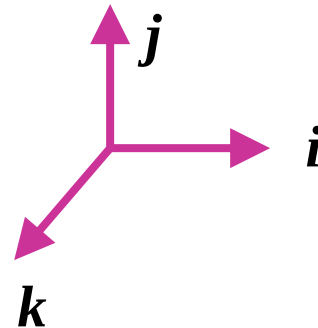
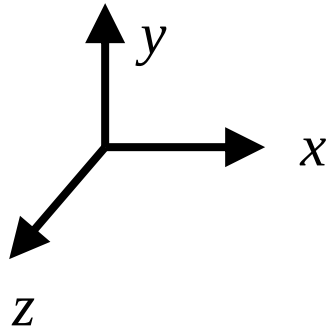
Mechanics: Scalars and Vectors

- A Vector **V** can be written as: $\mathbf{V} = V\mathbf{n}$

V = magnitude of **V**

$\mathbf{n}$ = unit vector whose magnitude is one and whose direction coincides with that of **V**

- Unit vector can be formed by dividing any vector, such as the geometric position vector, by its length or magnitude
- Vectors represented by Bold and Non-Italic letters (**V**)
- Magnitude of vectors represented by Non-Bold, Italic letters (V)

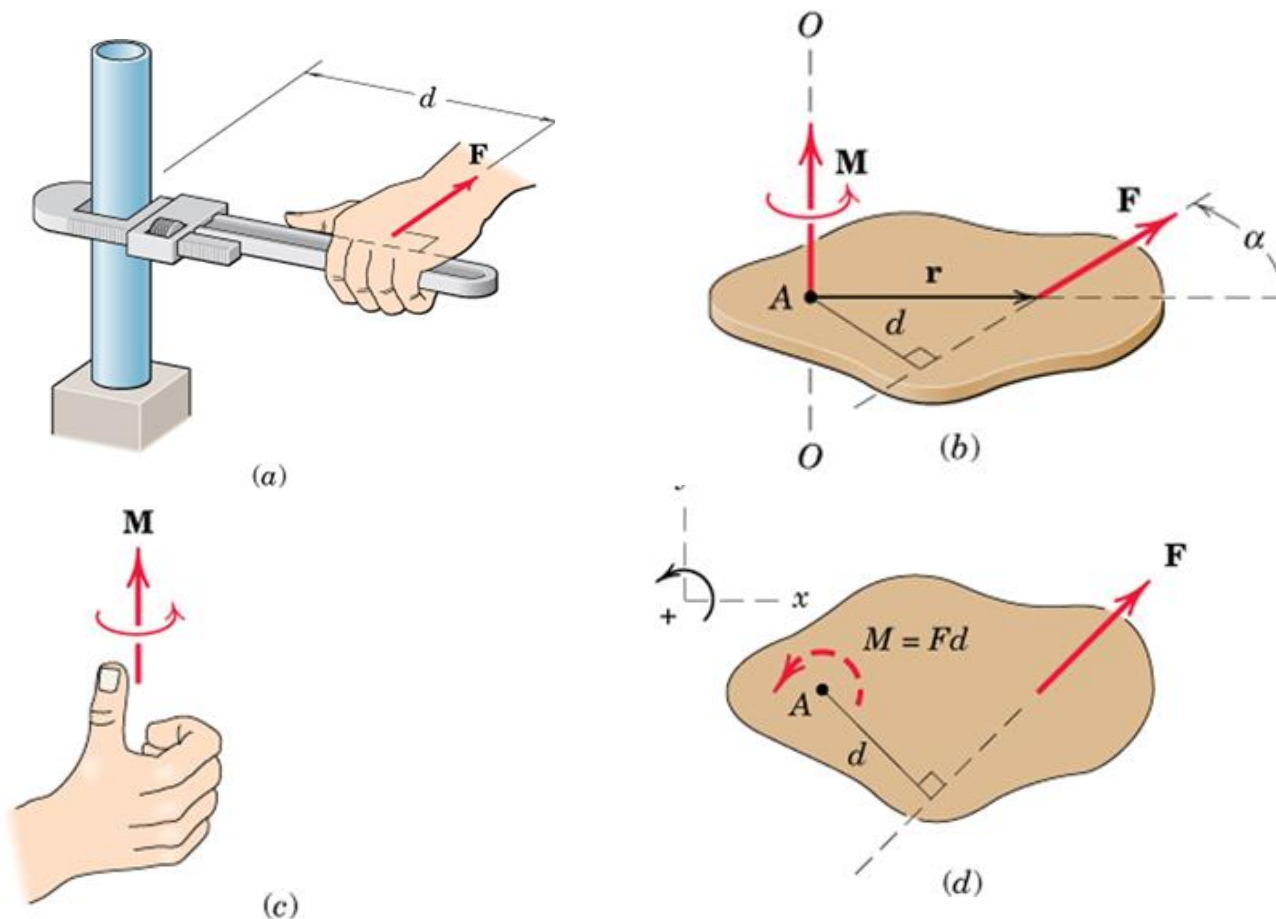


Moment of a Force (Torque)

- **Moment of a Force (F) @ point A**

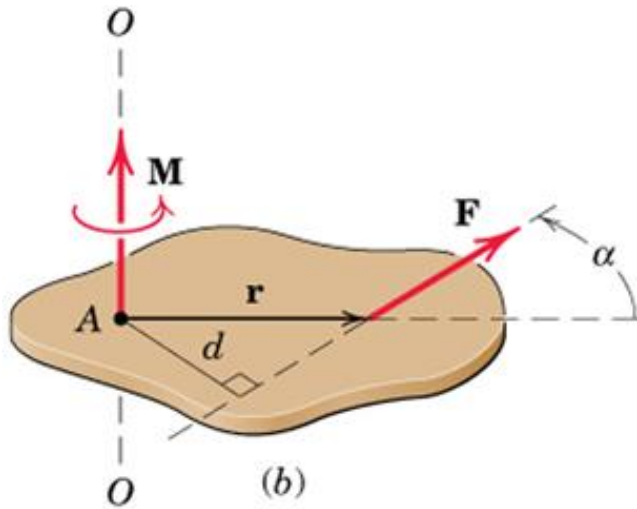
$$\mathbf{M}_O = \mathbf{r} \times \mathbf{F}$$

$\mathbf{r}$ = position vector directed from O to any point on the line of action of $\mathbf{F}$



Moment of a Force

- ❖ **F** tends to rotate the body about an axis along **M_O**



$$M_O = rF \sin \alpha = F(r \sin \alpha) = Fd$$

- ❖ Moment arm – $d = r \sin \alpha$

End